Graphics & Visualization

Chapter 18

Scientific Visualization Algorithms

Introduction

- Choice of visualization algorithm to be applied depends on:
  - *Type* of data
  - Desired visual *effect*
- Example:
  - Given a large scalar data set which must be displayed in its entirety → ray-casting or splatting algorithms
  - To examine areas of equal value more closely → marching cubes algorithm

Introduction (2)

- *Visualization object*: a function $V(S)$
  - Is the output of the *Data Preparation* stage (Chapter 10)
  - Domain S: space in which the experiment or simulation took place
    - E.g. 1: set of structured points in a 1-, 2-, 3-, or higher-D space; usually referred to as *grid* (most common domain type)
    - E.g. 2: regions of a continuous space
    - E.g. 3: enumerated set
  - Often, the domain will contain a time variable

Introduction (3)

- Range $V(S)$: data items produced by experiment or simulation for elements of the domain S
  - *Type* of range items of $V(S)$ distinguishes between visualization methods
  - Common range types: *scalar*, *vector*, *tensor*
- O notation

$$O : \text{domtype1} \times \text{domtype2} \times \ldots \times \text{domtypeN} \rightarrow \text{rangetype}$$

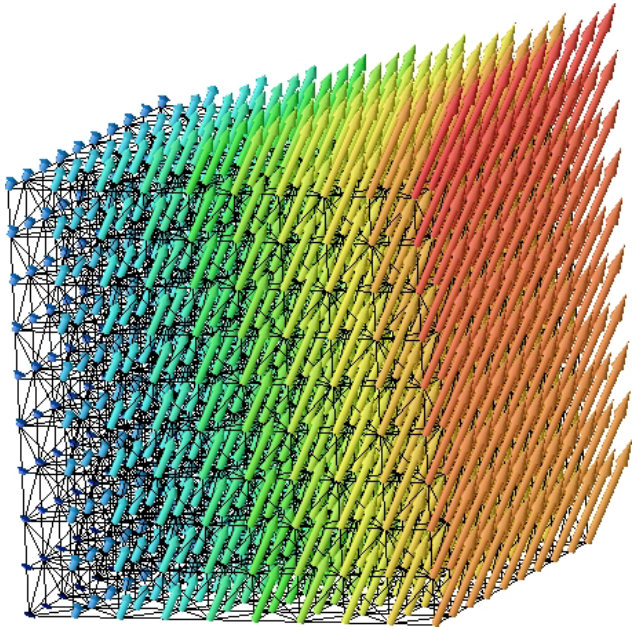
- Example
  - Visualization object that represents 2-element vector values (range) on a 3-D grid plus time (domain) has type:

$$X \times Y \times Z \times T \rightarrow \text{vector2}$$

  - Abbreviation: $O_{\text{domain_type}}^{\text{range_type}}$
    - So, a 3-element vector field over a 3-D grid is $O_{X \times Y \times Z}^{\text{vector3}}$

Introduction (4)

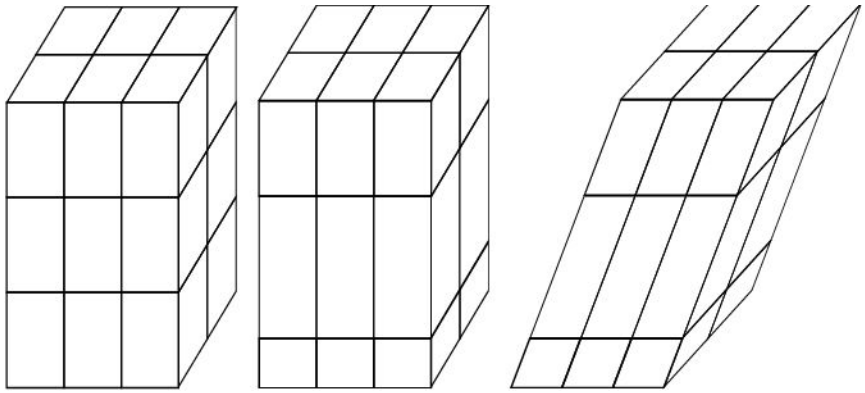
- An $O_{X \times Y \times Z}^{\text{vector3}}$:



- Consider the domain of 3D discrete space $X \times Y \times Z$ as a grid:
  - *Regular* → elementary volume elements are cubes of the same size
  - *Rectilinear* → elements are orthogonal parallelepipeds
  - *Structured* → elements are general parallelepipeds

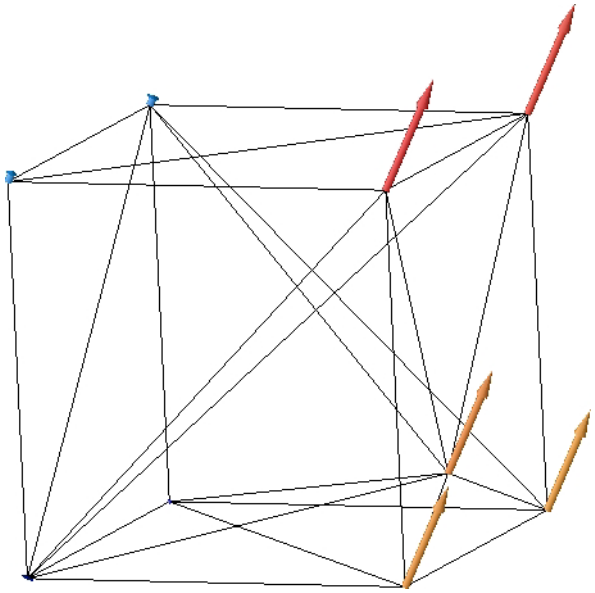
Introduction (5)

- Regular, rectilinear, and structured grids:



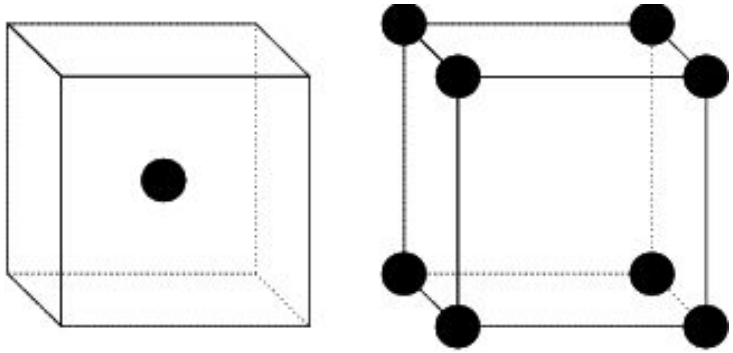
- Alternative:

  tetrahedral volume elements:



Introduction (6)

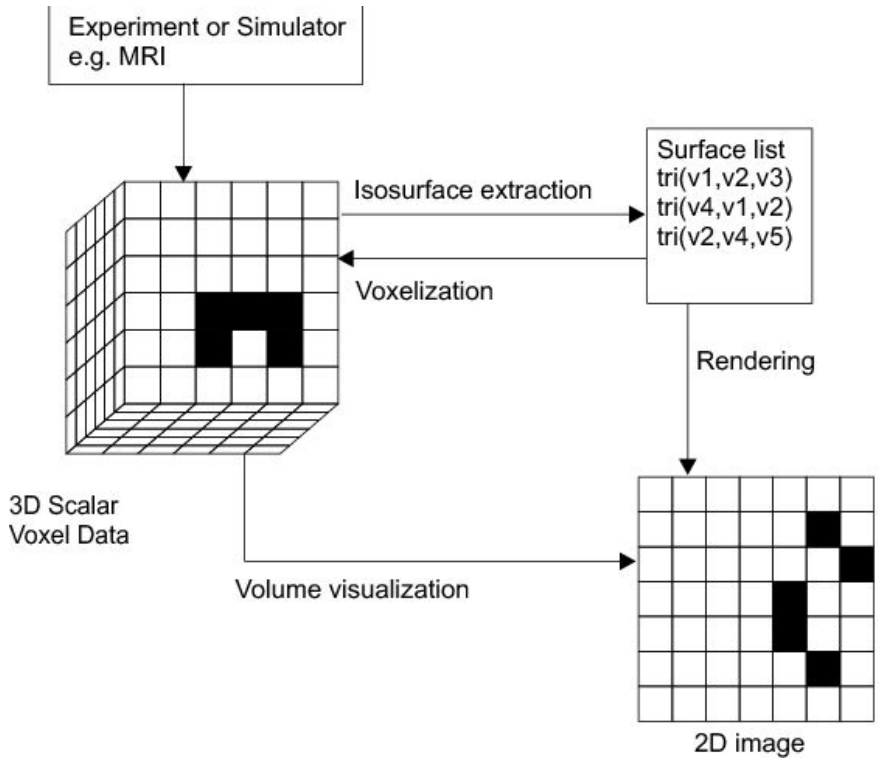
- Range values can be mapped onto the grid domain in 2 ways:
  - Associated with entire volume elements (*voxels*)
  - Associated with grid vertices (*cells*)



- Arbitrary viewing parameters
  - We assume a *discrete representation of continuous* data
  - Require field values at *arbitrary* 3D points
- To determine the value at an arbitrary 3D point, we have 2 options corresponding to the above mappings:
  - The point takes the constant value of the voxel that it belongs to
  - Interpolate from the vertex values of the appropriate cell (better)

Scalar Data Visualization

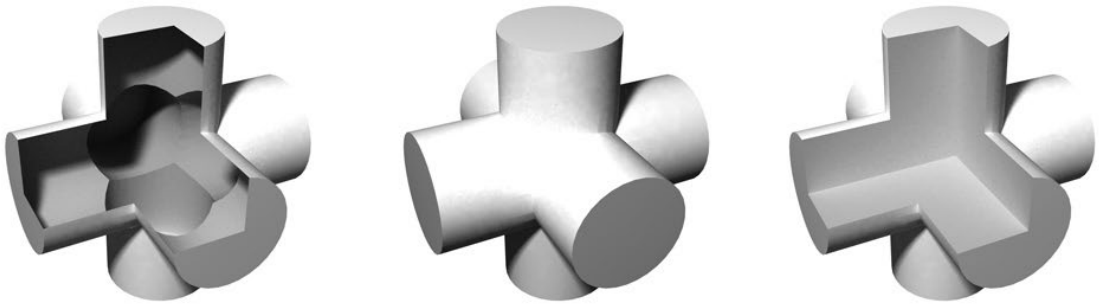
- Two main approaches to visualizing scalar data represented on a grid:



- S1. To observe one or more surfaces of constant value (*isosurfaces*) within the field → employ isosurface extraction algorithms

Scalar Data Visualization (2)

- Isosurfaces create sharp renderings & by transforming to a standard representation, they take advantage of widely available graphics techniques to accelerate rendering
- However, only part of the information present in the scalar field is visible on the isosurfaces



- S2. Display the entire field by employing a direct volume-visualization technique:
  - Such techniques are slow & generally result in blurry images
- The choice depends largely on the specifics of the application

Isosurface Extraction Algorithms

- Often data contain clusters of values which can be separated by surfaces
- Isosurface algorithms determine these separating surfaces after the user inputs one or more *isosurface value(s)*
- Creating isosurfaces was not straightforward until *marching cubes*
- Once these isosurfaces are established:
  - Quick and easy to display them with standard graphics techniques, as they consist of polygons

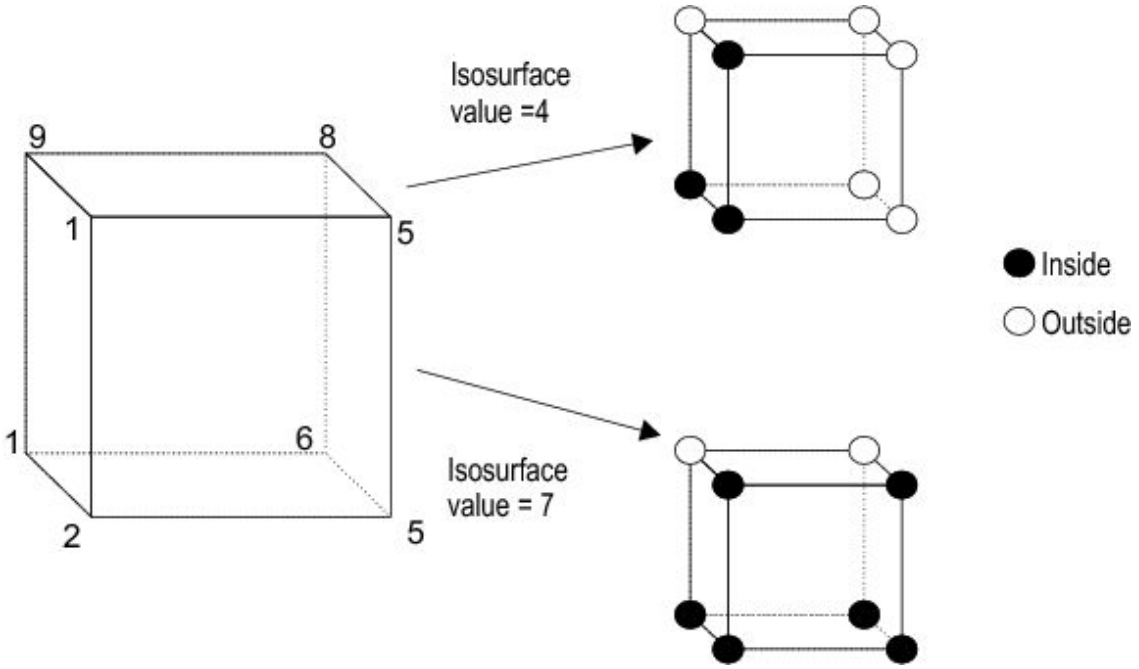
- Marching Cubes & Splitting Box algorithms

Marching Cubes Algorithm

- Input: Scalar volume data set $O^{\text{scalar}}_{X \times Y \times Z}$ and isosurface scalar value
- Output: list of polygons representing the isosurface
- Marching Cubes (MC) visits every *cube* of the volume data set
- For each cube, the field values at its 8 vertices are compared to the user-provided isosurface value
- Vertices are thus labeled as 1 (*inside*, smaller than isosurface value) or 0 (*outside*, greater than isosurface value)
- Vertex labels are then systematically concatenated & used as an index to a list of pre-computed surface-cube intersections

Marching Cubes Algorithm (2)

- Vertex labeling:



Marching Cubes Algorithm (3)

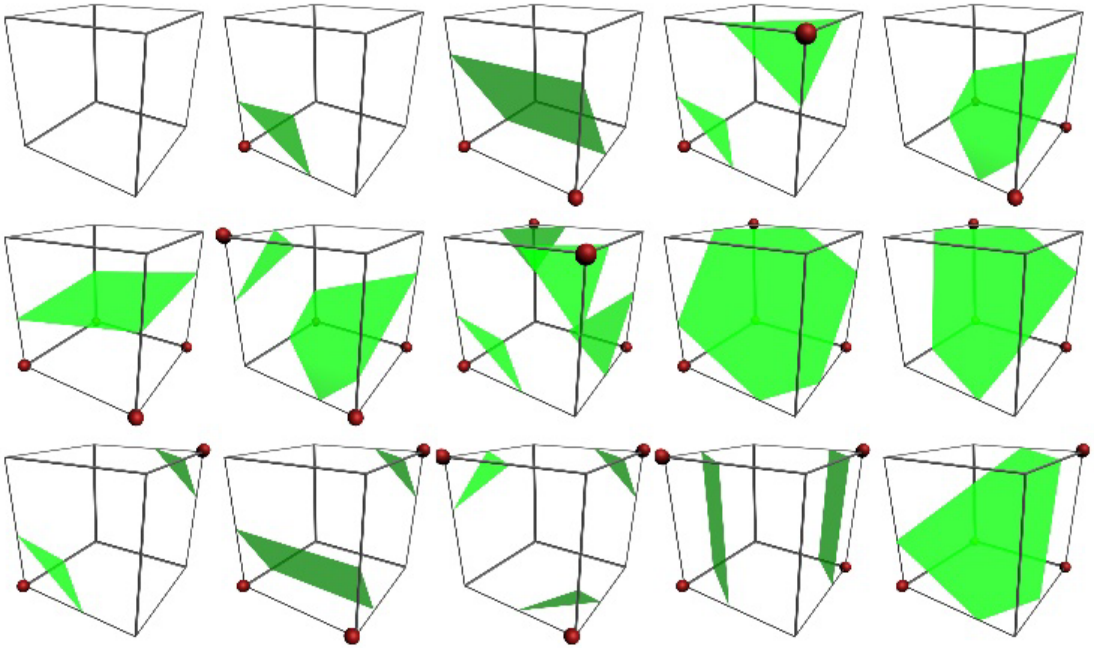
```
Void MC() {
    For (i= 0; i<maxcubeI; i++)
      For (j= 0; j<maxcubeJ; j++)
        For (k= 0; k<maxcubeK; k++) {
            // process cube (i,j,k)
            // label vertices as inside (1) or outside (0)
            l1=get_label (i,j,k);
            l2=get_label (i+1,j,k);
            ...
            l8=get_label (i+1,j+1,k+1);
            // concatenate the 8 labels (++ stands for the
            // string concatenation operator)
            index=l1++l2++l3++l4++l5++l6++l7++l8;
            // map index to one of the 15 basic cases
            // (symmetries) and get required transform
            bindex=map_2_basic_index(index);
            transform=map_2_basic_trans(index);
```

Marching Cubes Algorithm (4)

```
            // use bindex to select the appropriate
            // precomputed surface-cube intersection
            // and reverse transform it
            surface_list=
            precomputed_surfaces(bindex,transform^{-1});
            // use interpolation to place the
            // intersection surface precisely
            for (p=0; p<num_vertices(surface_list); p++)
                    compute_precise_edge_position(p,
                    cube_field_values(i,j,k));
            // calculate normals at intersection
            // surface vertices for rendering
            for (p=0; p<num_vertices(surface_list) p++)
                    compute_normal(p, cube_field_values(i,j,k));
        }
}
```

Marching Cubes Algorithm (5)

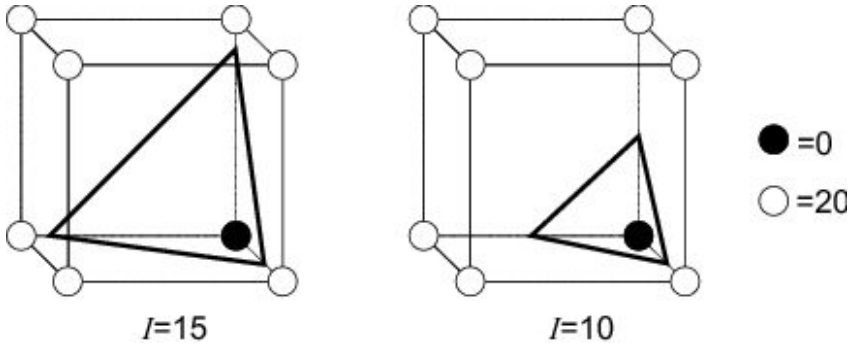
- 2^8 ways to label vertices of a cube:
  - Requires 256 pre-computed surface-cube intersection patterns
  - Reduced to just 15 by taking advantage of:
    - Mirror symmetry
    - Rotational symmetry
    - Inside/outside symmetry



Marching Cubes Algorithm (6)

- Each of the 15 intersection patterns provides the topology of the polygonal intersection surface *with respect to the cube edges*
- Symmetries used to go from the actual intersection pattern to one of the 15 basic cases form the transform for a cube
- The exact points of intersection along each cube edge are determined by linear interpolation, assuming linearly varying field:
  - If the edge vertices have associated field values v & v' & the isosurface value is I ($v < I < v'$) → intersection point p can be expressed as:

$$p = \frac{I - v}{v' - v}$$



Marching Cubes Algorithm (7)

- For realistic illumination, normal vectors of the isosurface on the vertices of isosurface polygons are must be computed
  - Classical way is to weigh the normals of neighbouring polygons (see illumination Chapter), but it is simpler, faster and more accurate to:
    1. Compute the gradient vectors of the scalar field at cube vertices

$$g_x(i,j,k) = \frac{v(i+1,j,k) - v(i-1,j,k)}{\Delta x},$$

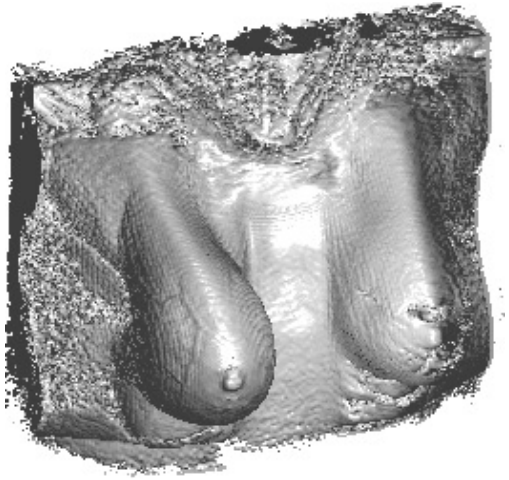
$$g_y(i,j,k) = \frac{v(i,j+1,k) - v(i,j-1,k)}{\Delta y},$$

$$g_z(i,j,k) = \frac{v(i,j,k+1) - v(i,j,k-1)}{\Delta z}$$

    2. Interpolate gradient vectors along cube edges & onto the vertices of the polygons

where $v(i,j,k)$ & $g(i,j,k)$ are the field value & gradient vector at cube vertex (i,j,k) and Δx, Δy, Δz are the differences in the x-, y-, & z-coordinates of the cube vertices involved

Marching Cubes Algorithm (8)



- MC can be improved in a number of ways:
  - E.g. avoid re-computation for common edges of neighboring cubes
- Major disadvantages of MC algorithm:
  - Large # of polygons created for the isosurface
  - This # is not proportional to the isosurface complexity:
    - Depends primarily on the density of the grid
- MC can be fully accelerated by the GPU

Splitting Box Algorithm

- Splitting box (SB) algorithm also creates an isosurface from volumetric scalar data sets:
  - Creates smaller # of polygons than MC by recursively subdividing the original volume **only until** the resulting elements possess a certain complexity property
- Definitions:
  - *Box*: Rectangular parallelepiped with edges parallel to the main axes of the grid
  - *Length* of an edge: # of grid vertices it contains
  - An edge has the *MC property* if it contains at most one isosurface transition
    - SB algorithm uses this generalized property to end the subdivision process as early as possible in the box hierarchy

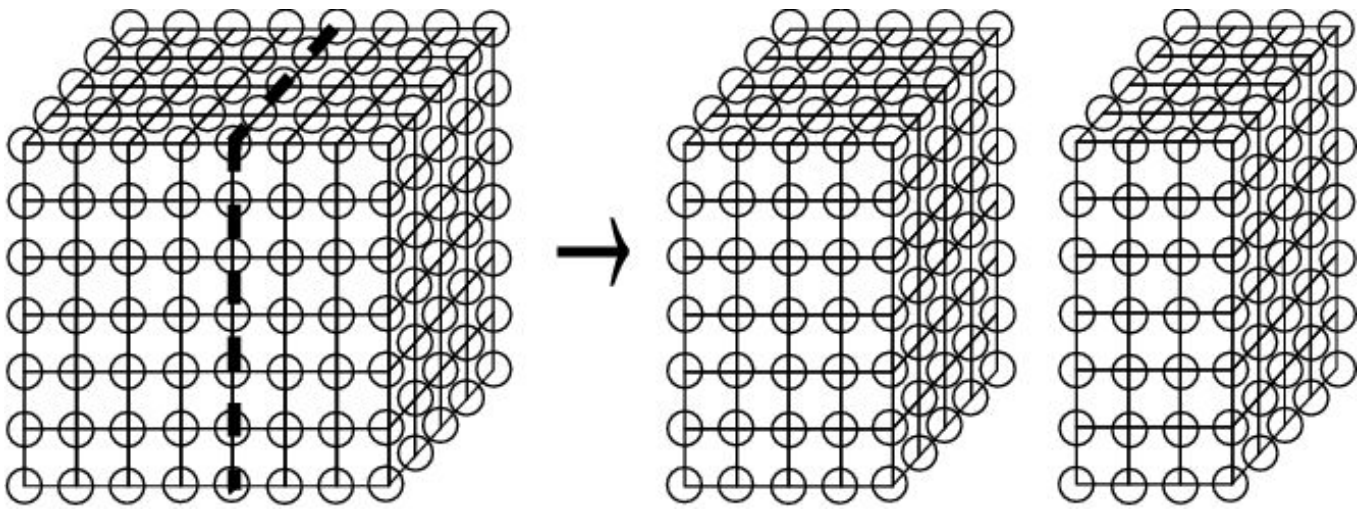
Splitting Box Algorithm (2)

- A box Face is MC if its 4 edges are MC
- A Box is MC if its 6 faces (or twelve edges) are all MC

```
void SB (box); {
      if  MC_property(box)
         generate polygons using the 15 cases of MC;
      else if  size(box)=2^3
         generate polygons by analytical processing;
      else {
         subdivide box along longest edge into box1 and box2;
         SB(box1);
         SB(box2);
      }
   }
```

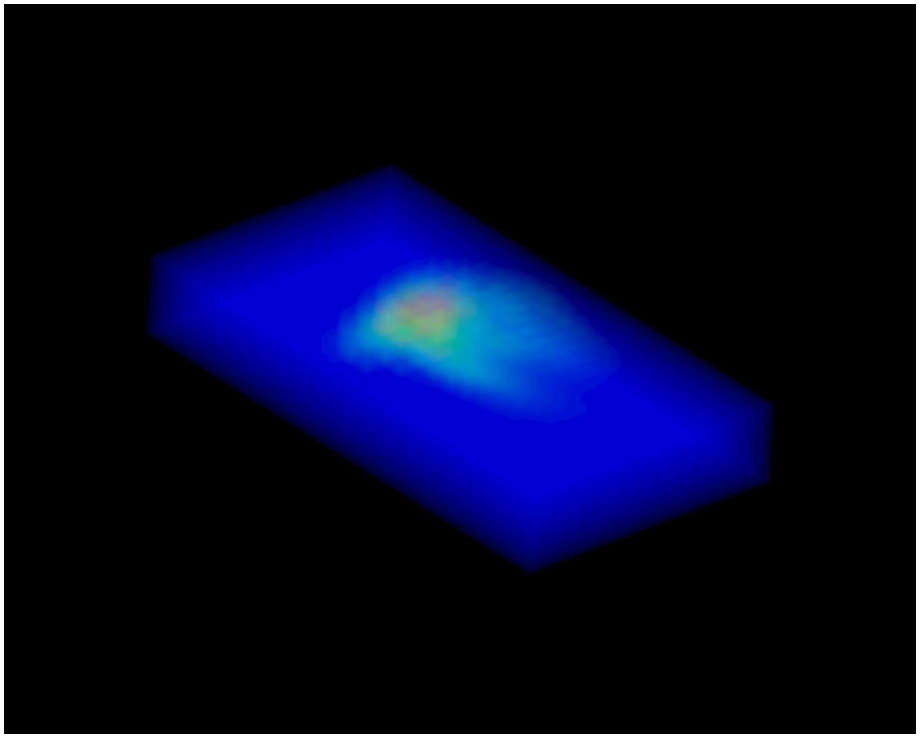
Splitting Box Algorithm (3)

- Box subdivision:



Direct Volume Visualization

- 3-D scalar data sets consisting of sampled data or representing amorphous phenomena are hard to represent using surfaces
- In such cases we can employ visualization algorithms that display the data by directly interrogating the data set
- Volume rendering:



Direct Volume Visualization (2)

- Two types of algorithms
  - *Backward projection* (*ray casting*):
    - Fires rays for each image pixel into the data set, obtains samples and combines them into a final color
  - *Forward projection* (*splatting*):
    - Projects each voxel in the data set onto the image plane and establishes which pixels it affects using a filter

Ray Casting

- Ray casting consists of 3 steps:
  - *Classify* each voxel according to its content
  - *Transform* rays or data so that they are aligned with the viewing direction
  - *Combine* the result along each ray
- **<u>Classification step:</u>**
  - Classifies each voxel depending on its material content
  - Result is a color and transparency value
- **Example:**
  - In a medical scanning application color & transparency values are assigned according to the x-ray absorption values
  - Set of possible materials: material = { air, fat, soft-tissue, bone}
  - Material i has an a-priori given probability distribution $P_i(I)$ to have intensity value I in homogeneous form
  - Which material(s) does a voxel contain & in what proportion?

Ray Casting (2)

- If $P(I)$ denotes the probability that a voxel has intensity value I:

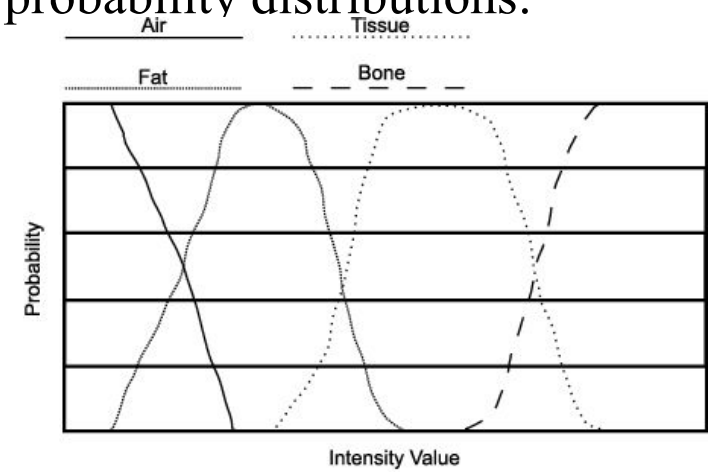
$$P(I) = \sum_{i=1}^{m} \rho_i P_i(I)$$

  where ρ_i is the proportion of material i in the voxel
- Bayesian estimate of amount of material i in voxel with intensity I:

$$\rho_i(I) = \frac{P_i(I)}{\sum_{j=1}^{m} P_j(I)}, \qquad \sum_{j=1}^{m} \rho_j(I) = 1$$

- Color/transparency C of the voxel computed as: $C = \sum_{i=1}^{m} \rho_i C_i$

  where $C_i = (\alpha_i R_i, \alpha_i G_i, \alpha_i B_i, \alpha_i)$ is the color and transparency value that corresponds to material i
- Material intensity probability distributions:

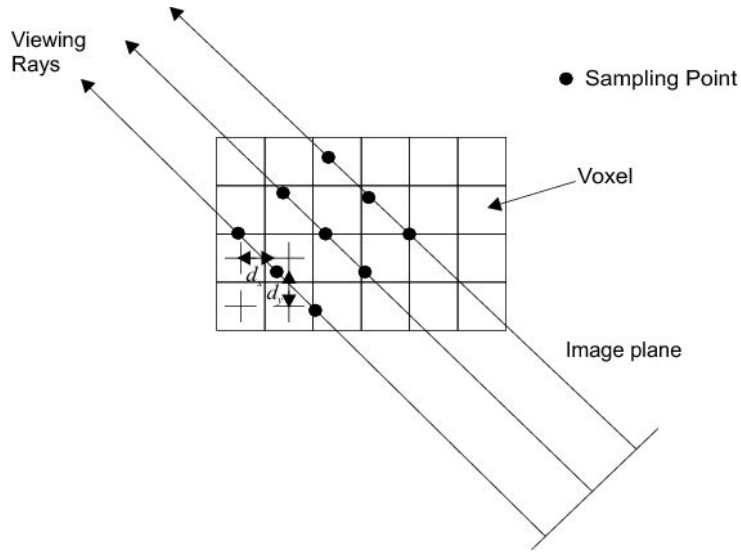


Ray Casting (3)

- **Transformation Step:**
  - Two ways of aligning to the viewing direction:
    - *Ray transformation*
    - *Data transformation*

Ray Transformation:

- Casts rays into the volume data & takes samples at equidistant points along each ray



- Samples are computed by tri-linear interpolation from the 8 nearest voxel values

Ray Casting (4)

- **Example:** $v_{i,j,k} \mid i,j,k \in \{-,+\}$ represent the values of the 8 surrounding voxels for a sampling point s & d_x^-, d_y^-, d_z^- represent the distances from s to the centers of the 3 voxels with the smaller indices in each axis as a portion of the inter-voxel distance; the interpolated value at s is:

$$v_s = (1-d_x^-)\left[(1-d_y^-)\left[(1-d_z^-)v_{---} + d_z^- v_{--+}\right] + d_y^-\left[(1-d_z^-)v_{-+-} + d_z^- v_{-++}\right]\right]$$
$$+d_x^-\left[(1-d_y^-)\left[(1-d_z^-)v_{+--} + d_z^- v_{+-+}\right] + d_y^-\left[(1-d_z^-)v_{++-} + d_z^- v_{+++}\right]\right]$$

***Data Transformation:**

- Aligns the volume data with the viewing direction
- If z is the viewing axis → voxel data are realigned so that voxels with the same z-coordinate lie on the same viewing ray
- Data transformation achieved by a shear-warp operation
- Slices of the volume data are sheared in xy-plane by factors s_x & s_y
- Data of each slice are re-sampled using bi-linear interpolation

Ray Casting (5)

- Shear matrix for a parallel projection is:

$$SH_{\text{par}} = SH_{xy}(s_x, s_y) = \begin{bmatrix} 1 & 0 & s_x & 0 \\ 0 & 1 & s_y & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

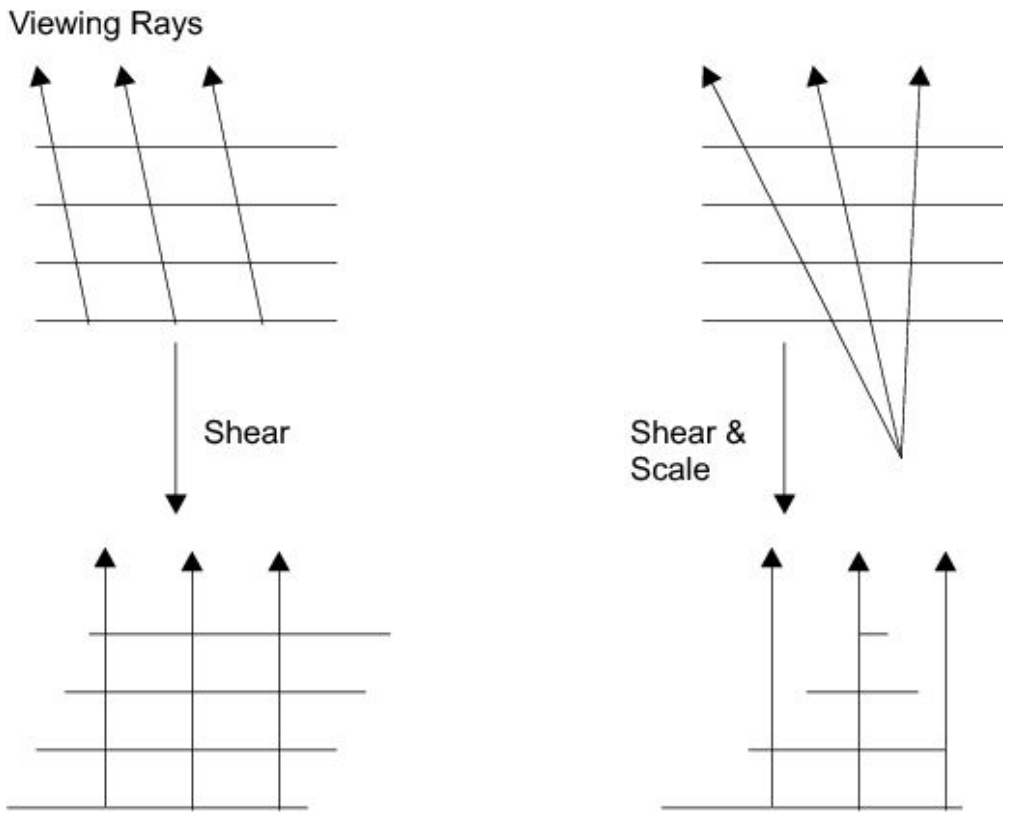
- In the case of a perspective projection also need to scale each slice → scale factor s is determined from the viewing transformation:

$$SH_{\text{pers}} = \begin{bmatrix} 1 & 0 & s_{x'} & 0 \\ 0 & 1 & s_{y'} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & s & 1 \end{bmatrix}$$

  - A volume data slice at $z=z_0$ is thus scaled by $1/(1+s * z_0)$ to return to normal homogeneous form
  - <u>Problem</u>: Non-homogeneous sampling of volume data caused by the diverging rays (sampling density is a function of depth)

Ray Casting (6)

- <u>Solution:</u> Adaptively introduce extra rays as a function of depth
  - Rays then get sub-sampled in slices → end up with initial ray resolution
- Shear operation for parallel (left) & perspective (right) projection:



Ray Casting (7)

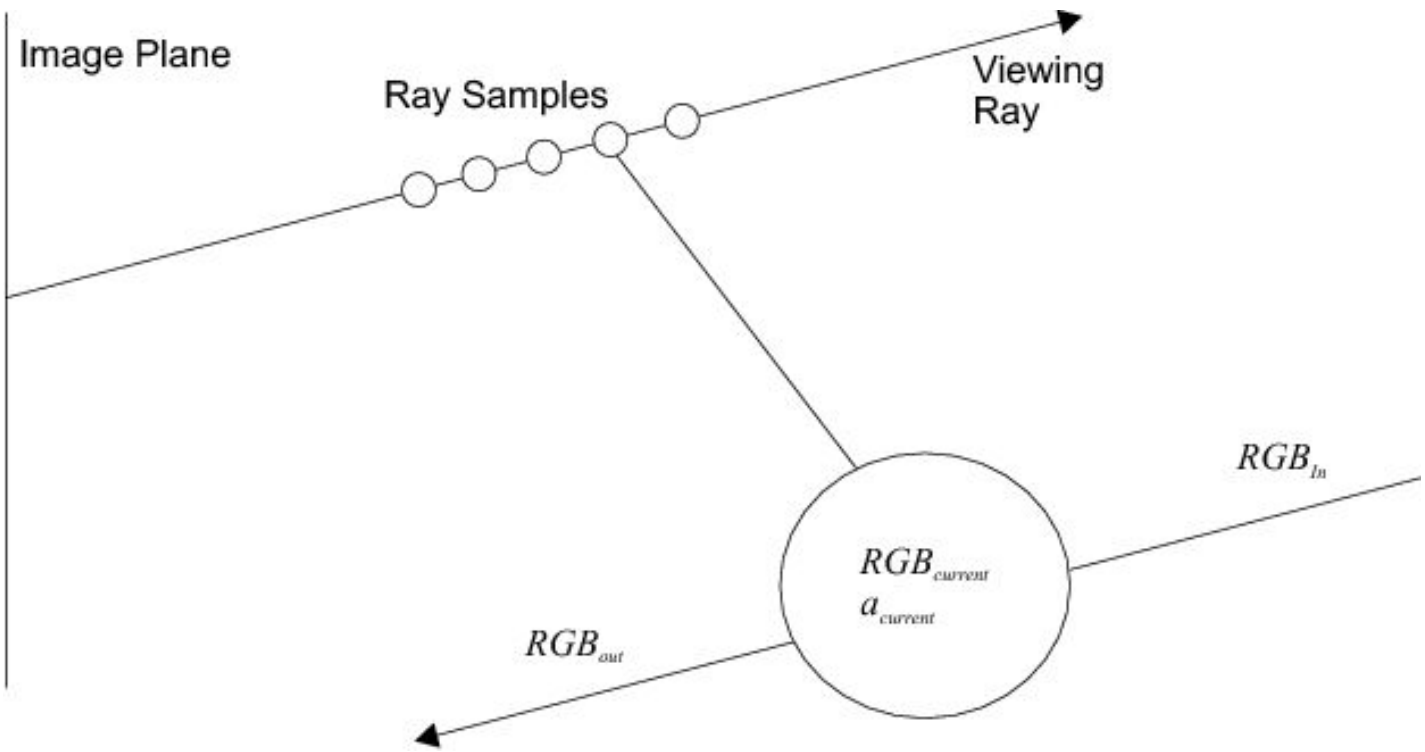
- **Combination Step:**
  - Determines the resulting color value along each ray by:
    - combining the value at successive sample points (ray transformation) or
    - successive voxels along the *Z*-direction (data transformation)
  - Both cases are referred to as *ray samples*
  - Value at each ray sample stored in RGBA form → traverse ray samples from *back-to-front*
  - At each ray sample compute outgoing color value RGB_{out} as a function of incoming value RGB_{in}, and the color & transparency properties of current ray sample:

$$RGB_{\text{out}} = RGB_{\text{in}}(1-\alpha_{\text{current}}) + RGB_{\text{current}}\alpha_{\text{current}}$$

where $\alpha_{current}$ ranges from 0 (totally transparent) to 1 (totally opaque)

Ray Casting (8)

- Combining the ray samples:



Ray Casting (9)

- To allow for early ray termination, we can reverse the process to *front-to-back*
  - Must accumulate color and transparency values

$$RGB_{\text{out}} = RGB_{\text{acc}} + (1-\alpha_{\text{acc}})RGB_{\text{current}}$$

$$\alpha_{\text{acc}} = \alpha_{\text{acc}} + (1-\alpha_{\text{acc}})\alpha_{\text{current}}$$

where RGB_{out} is the outgoing color in the front-to-back direction

*Splatting

- The splatting algorithm:
  - Gives voxels the priority
  - Considers each voxel's projection onto the image plane (i.e. the pixels)
  - Appears like the voxel was "thrust onto" the image plane
- The discrete voxel space represents a continuous volume
- Values *f (x, y, z)* can be reconstructed by:

$$f(x,y,z) = \sum_i \sum_j \sum_k v(i,j,k) \cdot h(x-i, y-j, z-k),$$

where *v(i, j, k):* the discrete voxel values

h: the reconstruction kernel

- The summation is taken over a 3D volume
  - Volume's size is equal to that of the reconstruction kernel

*Splatting (2)

- Consider the contribution contri*(x,y,z)* of a voxel *(i,j,k)* to a point *(x,y,z)*

$$\text{contri}(x, y, z) = v(i, j, k) \cdot h(x - i, y - j, z - k)$$

- Assume *z* to be perpendicular to the image plane:

$$\text{contri}(x, y) = \int_{-\infty}^{\infty} v(i, j, k) \cdot h(x - i, y - j, z) dz$$

- *v(i,j,k)* does not depend on *z* and can be taken outside the integral:

$$\text{contri}(x, y) = v(i, j, k) \cdot \int_{-\infty}^{\infty} h(x - i, y - j, z) dz$$

*Splatting (3)

- A kernel is centered at every voxel
- Its contributions to image-space pixels can be determined by projecting the kernel onto image space
- All kernels have the same projection, called **footprint:**

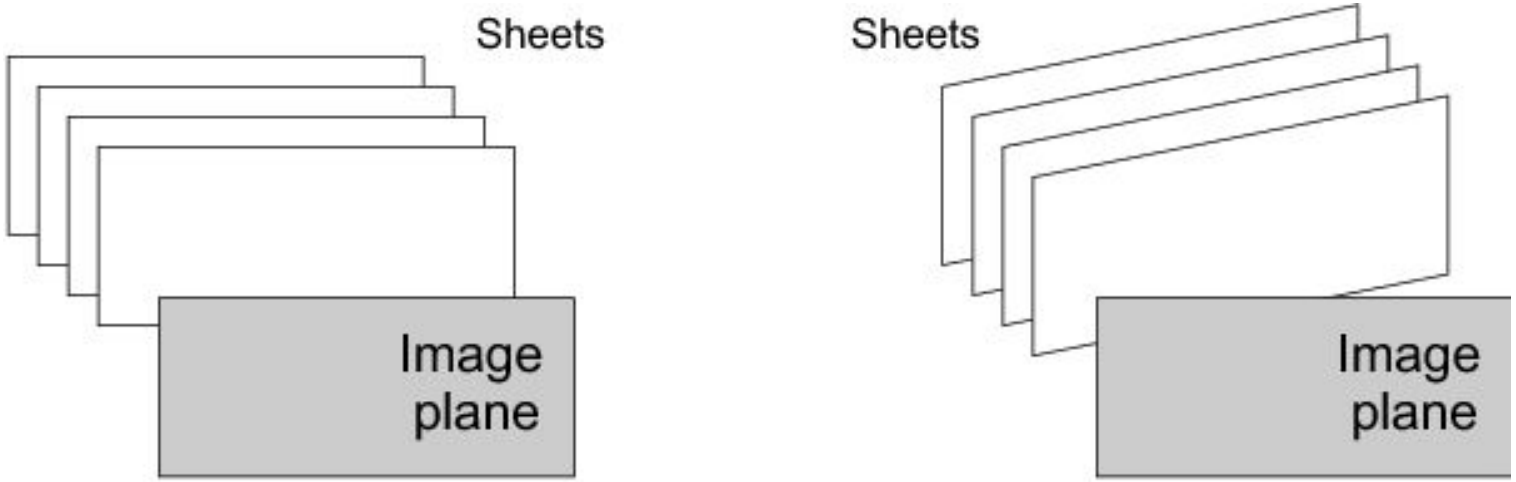
$$\text{footprint}(\alpha, \beta) = \int_{-\infty}^{\infty} h(\alpha, \beta, z) dz$$

where α, β: represent the image-space X- and Y-displacement from the central pixel of the kernel projection

- If the image plane is not aligned with the axes of the voxel volume, then the footprint function is slightly more complicated

*Splatting (4)

- The voxels are processed in **sheets**
- Sheet: a plane of voxels parallel to the image plane
- If the image plane is not aligned with the voxel space axes:
  - Define the sheets by the pair of voxel axes most parallel to the image plane



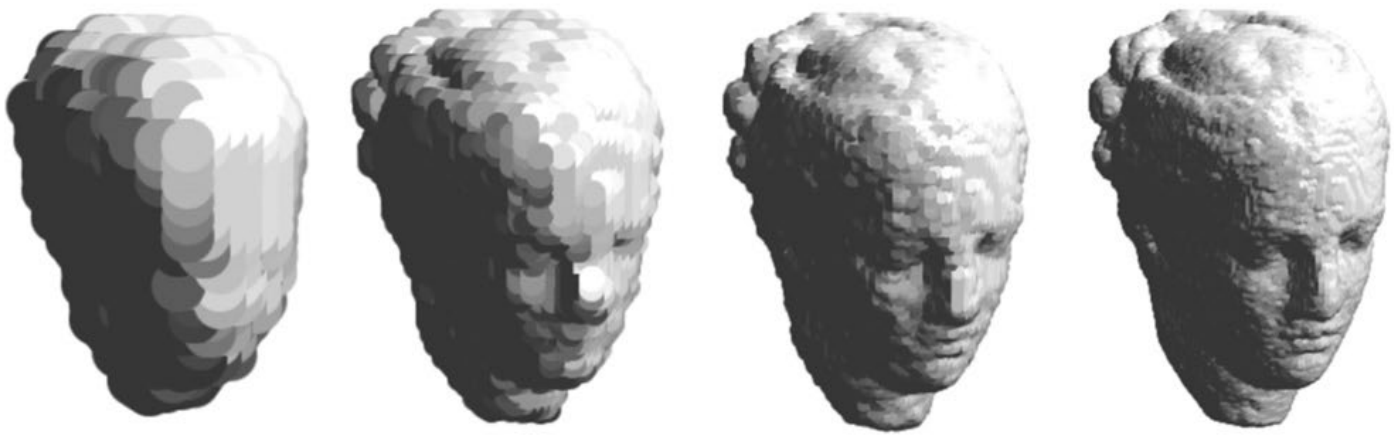
*Splatting (5)

- Algorithm: Processing starts with the sheet nearest to the observer
  - front-to-back: allows for early termination

```
for each sheet s front-to-back
        for each voxel (x,y,s) in sheet s
            for all footprint offsets (a,b) {
                frame_buffer(x+a,y+b)=
                    frame_buffer(x+a,y+b)+
                    voxel(x,y,s)*footprint(a,b)*
                    (1-transparency_buffer(x+a,y+b))
                transparency_buffer(x+a,y+b)=
                    min(1,transparency_buffer(x+a,y+b)+
                    transp(voxel(x,y,s))
                    *footprint(a,b))
            }
```

*Splatting (6)

- Splatting
  - Requires more computation as it processes the entire voxel space
  - Does not take advantage of bounding volumes
  - Voxels are processed independently
    - Is more amenable to parallel implementation than ray-casting
  - Very easy to integrate a multi-resolution representation of the volume data into the rendering stage



Vector Data Visualization

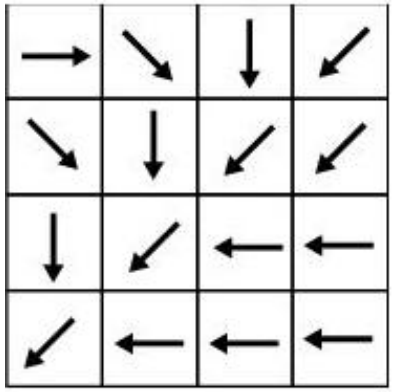
- Vector fields are common results of experiments and simulations
  - electromagnetic fields
  - derivatives of scalar fields
  - wind-velocity data
- Extra complexity as each element has several dimensions
- The dimensionality of the definition grid and that of the vectors in the field are independent

$$O_{X\times Y}^{\text{vector3}} \quad \textit{and} \quad O_{X\times Y\times Z}^{\text{vector2}} \ : \ \text{possible fields}$$

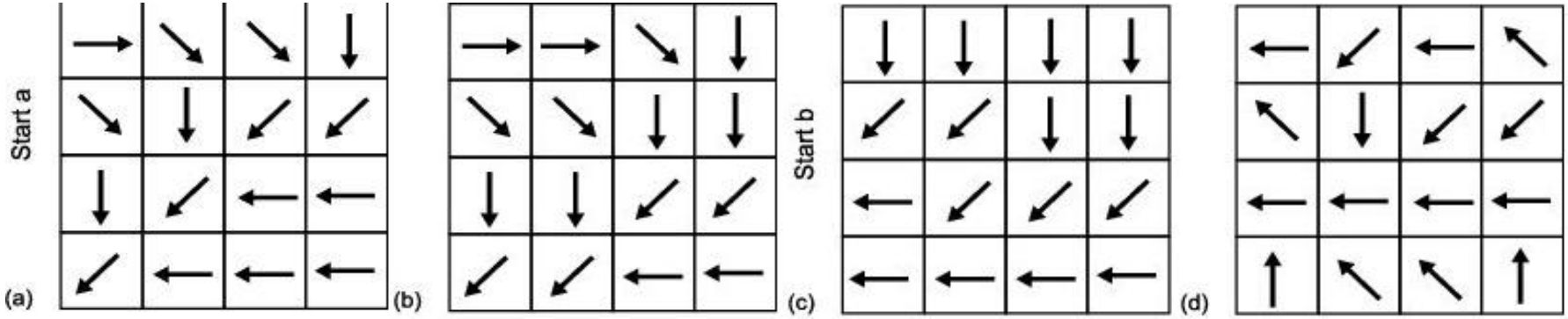
$$O_{X\times Y}^{\text{vector2}} \quad \textit{and} \quad O_{X\times Y\times Z}^{\text{vector3}} \ : \ \text{more frequently}$$

Vector Data Visualization (2)

- Static (or steady) vector fields
  - Do not change over time
  - E.g. the constant gravitational field between a set of static objects



- Dynamic (or unsteady, time-varying) vector fields
  - Represent a dynamic phenomenon
  - E.g. wind velocity and direction data during a day
  - Represented by "snapshots" of the field at discrete points in time
  - Each snapshot ≡ static vector field



Vector Data Visualization (3)

- Function types of 3D vector fields over 3D grids

$$V_{\text{static}} : O^{\text{vector3}}_{X \times Y \times Z},$$

$$V_{\text{dynamic}} : O^{\text{vector3}}_{X \times Y \times Z \times T}$$

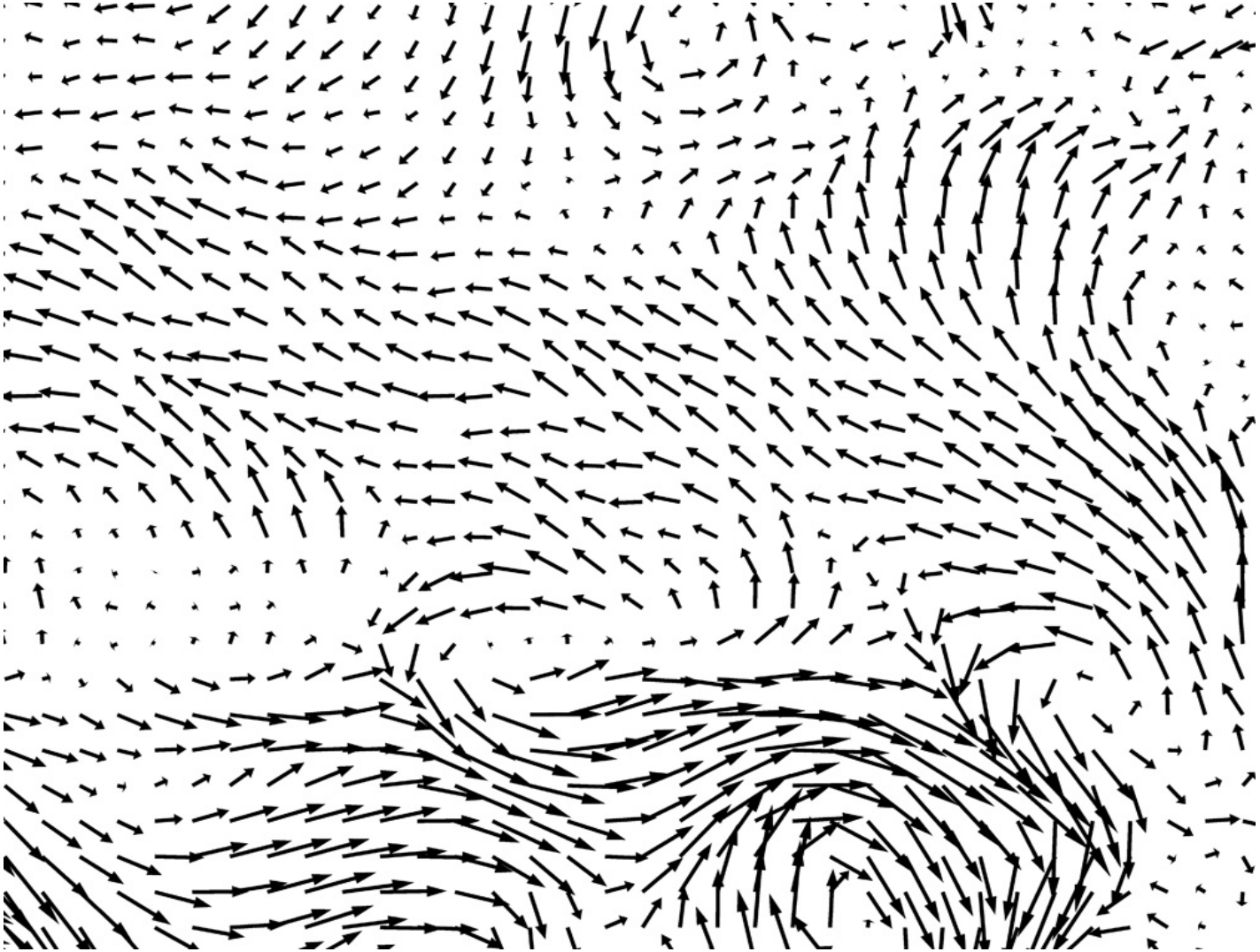
the last parameter in V_{dynamic} refers to time

Hedgehogs

- Representation of vectors as arrows
- Extensively used to represent vector fields
- Arrow length & direction → corresponding quantities of vector
- Dynamic vector fields can be represented by animations
  - Each frame is the arrow plot of the field at a specific time instant
- Major problems:
  - Visual clutter
    - In the case of dense fields
  - Projective distortion (foreshortening)
    - When higher dimensional fields are projected to 2D for display purposes
  - Discrete by nature
    - Original data is usually continuous and patterns such as spiraling & advection can be missed

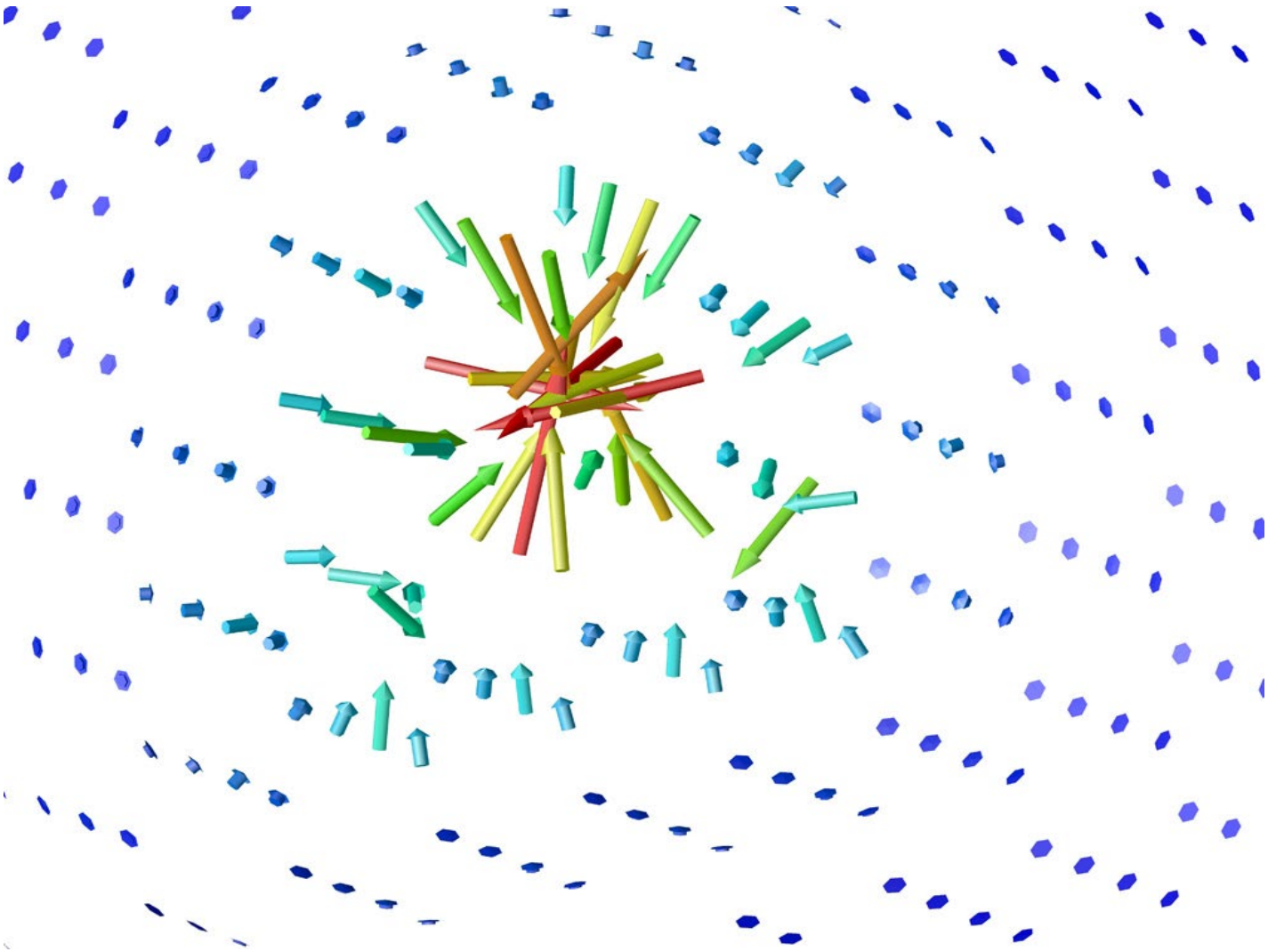
Hedgehogs (2)

Example $O_{X \times Y}^{\text{vector2}}$



Hedgehogs (3)

Example $O_{X \times Y \times Z}^{\text{vector3}}$



Particle Advection

- Imagine the trace of single particles through a vector field
  - think of a wind-tunnel experiment
  - we can release a small and light ping-pong ball and observe the path that it takes
  - better, release a small number of colored balls at different points and observe their joint behavior
- Rely on the existence of flow patterns in vector fields
- Display the effect of these flow patterns on weightless and frictionless particles that are advected through the field from their initial position

Particle Advection (2)

- **Visualization point** is a triplet:

$$\text{vispoint} : [x, y, z]$$

- **Visualization curve** is a set of connected points :

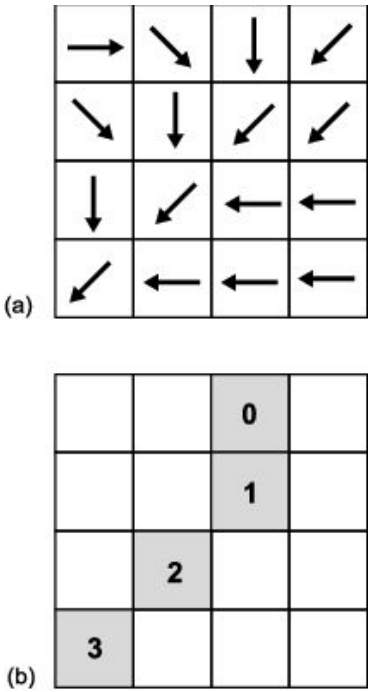
$$\text{viscurve} : \{\text{vispoint}\}$$

- **Streamline** is a function
  - takes a static vector field and a set of initial points
  - produces a set of visualization curves, one for each point:

$$\text{streamline} : V_{\text{static}} \times \{\text{vispoint}\} \rightarrow \{\text{viscurve}\}$$

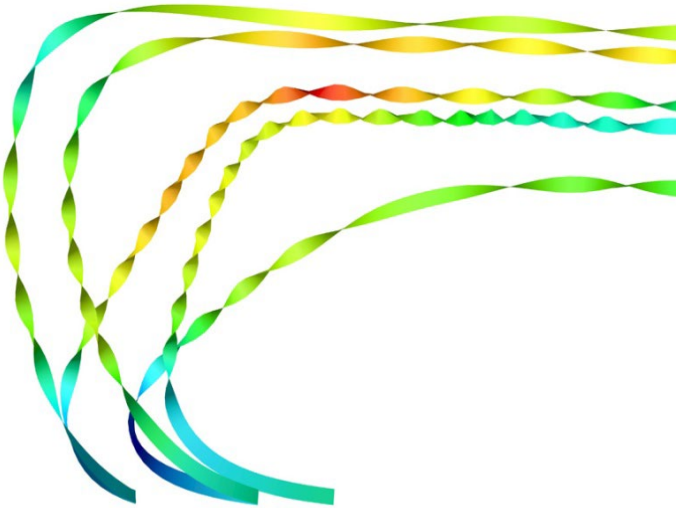
  - is the trace of a particle in a static vector field $\vec{S}$
- If π is the parameter of the streamline and **s** is a point on it:

$$\frac{d\mathbf{s}}{d\pi} = \vec{S}(\mathbf{s}(\pi))$$



Particle Advection (3)

- **Twist** of a vector field along streamlines
  - Visualized by plotting ribbons
  - **Ribbons** are the result of connecting the traces produced by pairs of neighboring particles
- Can plot an extra parameter by color coding its values along the length of the streamline



Particle Advection (4)

- Dynamic vector field $\vec{D}$: can use streamlines to visualize a single time instant t :

$$\frac{d\mathbf{s}}{d\pi} = \vec{D}(\mathbf{s}(\pi), t)$$

- Better alternatives exist for dynamic vector fields
  - Pathlines
  - Streaklines

Particle Advection (5)

- **Pathline**
  - A function: given a set of initial spacetime points produces the trace curve of each point through a dynamic field

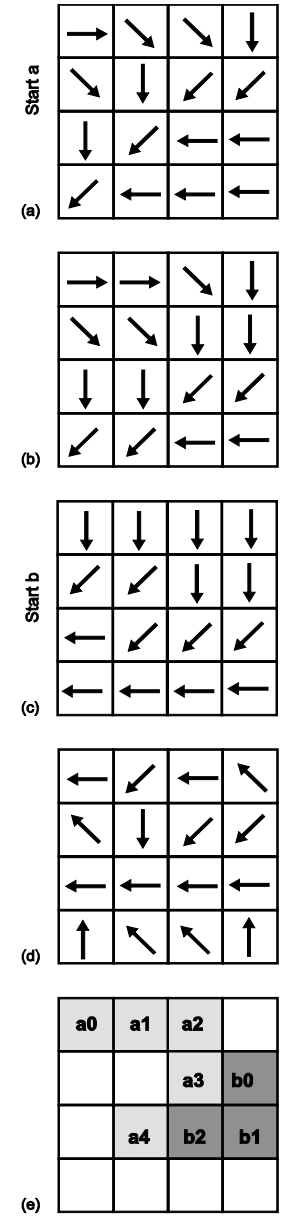
$$\text{pathline} : V_{\text{dynamic}} \times \{\text{initial}\} \rightarrow \{\text{viscurve}\}$$

where initial = (vispoint, time)

- Successively replaces the initial point by the result of advecting it through each instance of the field:

$$\frac{dp(\text{init},t)}{dt} = \vec{D}(p(\text{init},t),t)$$

where $p(\text{init}, t)$: the pathline starting at init=($\mathbf{s}_0$, t_0) at parametric time t



Particle Advection (6)

- **Streakline**
  - Dynamic curve, effect of injecting dye at given spacetime points in a dynamic field
  - Past curve positions are continuously advected by field
  - Can be viewed as a function that takes a dynamic vector field and a set of initial spacetime points and produces an equinumerous set of visualization curves, one for each point

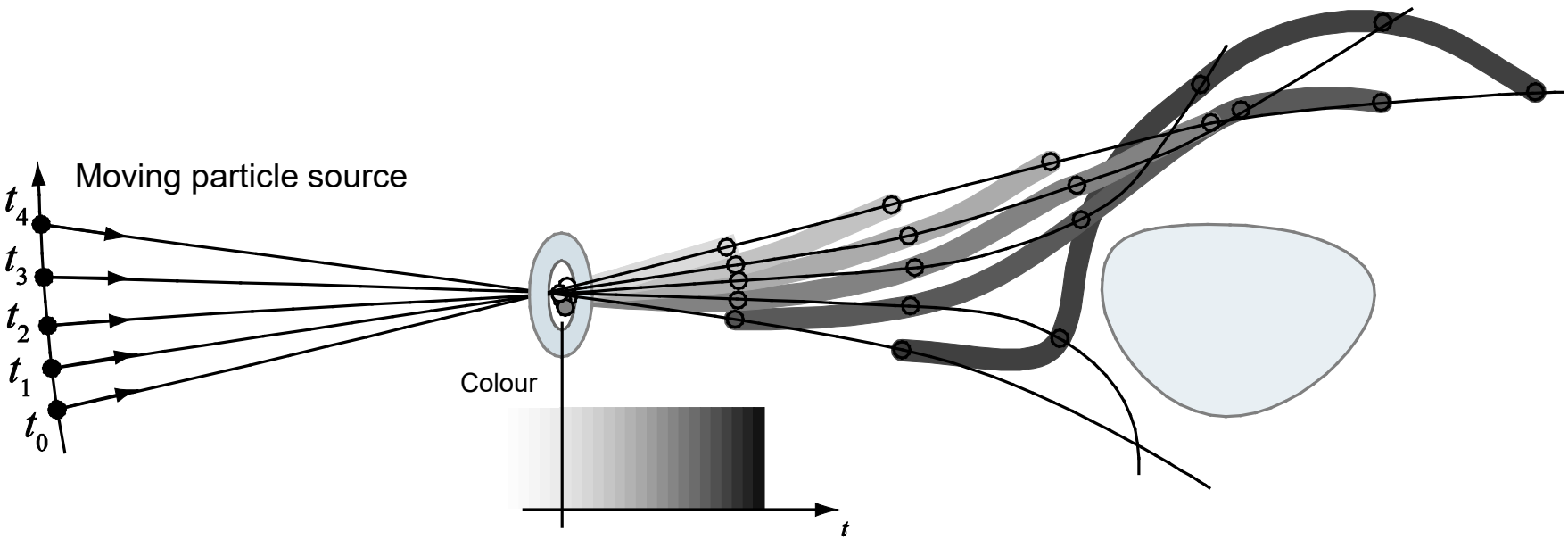
$$\text{pathline}: V_{\text{dynamic}} \times \{\text{initial}\} \rightarrow \{\text{viscurve}\}$$

where initial = (vispoint, time)

  - A streakline S from init=($\mathbf{s}_0, t_0$) at time t is S(init,t)=p(init,t): the endpoint of the streakline at time t coincides with that of a pathline
    - But all previous points of the streakline would have been advected (≠pathline)
  - For a static field: streamline ≡ pathline ≡ streakline

Particle Advection (7)

Streaklines



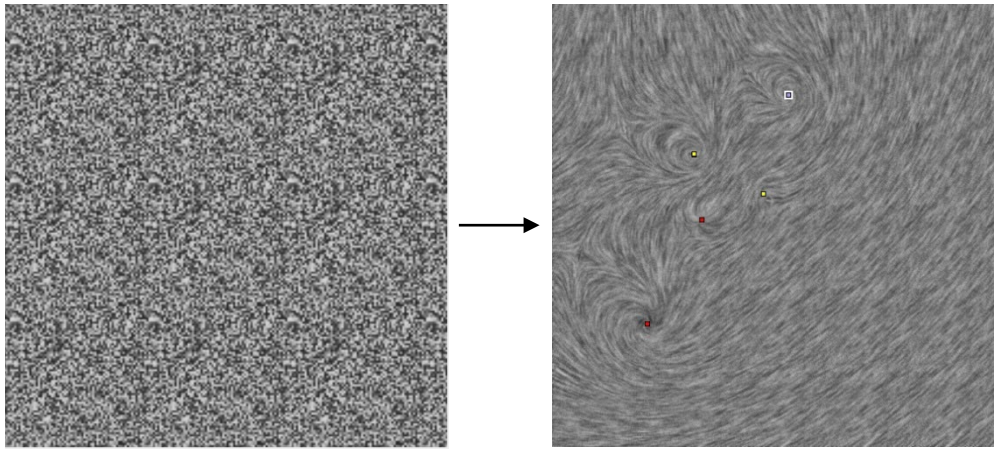
Texture Based Methods

- Allow the visualization of entire dense vector field
  - Via deformation on an input image, usually a noise image
- Line Intergral Convolution
- Image Based Flow Visualization

Line Integral Convolution

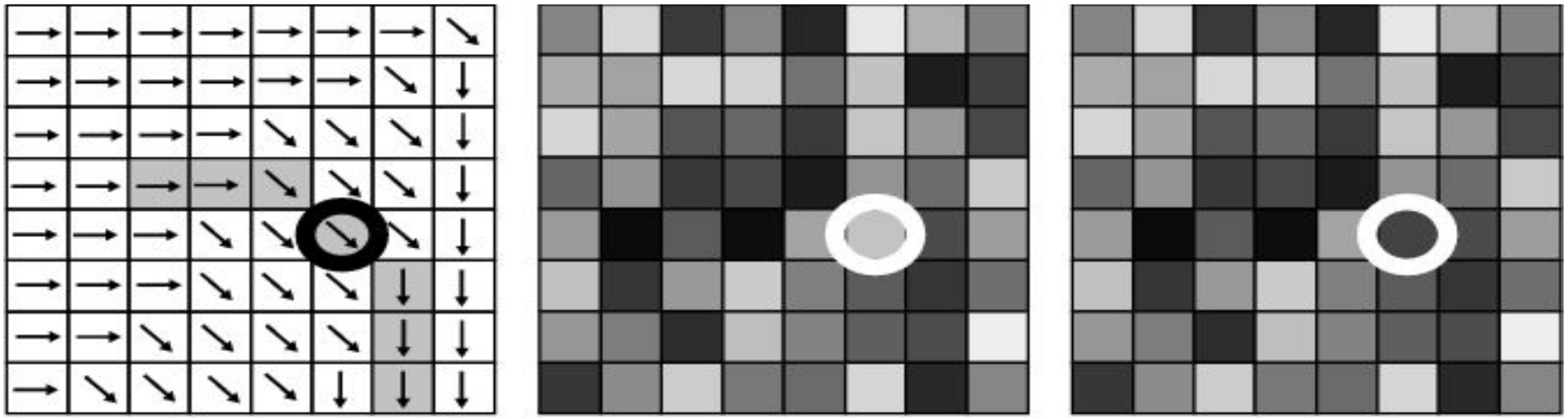
- Particle advection techniques present only a very small portion of the information and can lead to wrong conclusions about a field
- Line integral convolution (LIC)
  - Allows the global visualization of dense static vector fields over 2D or 3D grids
  - An input texture with resolution equal to the cell count of the grid is "locally" blurred to produce an output texture of the same size
- The vector field is visualized via its blurring effect on the texture
- The input texture may be
  - related to the vector field
  - completely unrelated,

  such as a noise image:



Line Integral Convolution (2)

- Assume an $O_{X \times Y}^{\text{vector2}}$ vector field
- The local behavior of the vector field can be approximated by computing a streamline that
  - starts at the center of a cell [pixel] **p**
  - extends in both the positive and the negative directions of the field
- The value at pixel **p** of the output image is computed by:
  - A convolution of a 1D strip of the input image with a 1D convolution filter
  - The convolution is performed over L cells of the streamline of **p** in each direction



Line Integral Convolution (3)

- If O and I are the output and input images, the LIC function is:

$$O(\mathbf{p}) = \sum_{\mathbf{k} \in A} I(\mathbf{k}) \cdot h(\mathbf{k})$$

where A: the set of cells of the streamline within a discrete cell distance L from **p** [shaded in the previous Figure (left)]

- L is half the length of the convolution kernel

$$h(\mathbf{k}) = \int_{\lambda_1}^{\lambda_2} k(w)dw$$

where

- λ_1 : the arclength of the streamline from **p** to the point where it enters cell **k**
- λ_2 : the arclength of the streamline from **p** to the point where it exits cell **k**
- $k(w)$: the convolution filter function

Line Integral Convolution (4)

- The length of the convolution kernel ($2L$) is a critical parameter
  - large L → LIC functions of most cells have similar values (more blurring)
  - small L→ insufficient filtering (no blurring)

Line Integral Convolution (5)

- So far LIC was used visualize vector **direction**
- To visualize **magnitude**
  - Vary L according to local vector magnitude
  - Amount of blurring is proportional to vector magnitude
- Generalization of LIC to $O^{\text{vector3}}_{X \times Y \times Z}$ fields:
  - Number of voxels in such 3D fields can be very large
  - Computational cost can be proportionally high
  - Hard to visualize such a dense 3D output
  - Define a 3D ROI by placing constraints on some scalar value of the field
  - Use these constraints to mask out (set to 0) parts of the 3D input texture
  - LIC calculations performed on voxels that correspond to non-zero input texture elements

Line Integral Convolution (6)

```
for each voxel v
    if(scalar_test(v)∉ ROI) set input_texture(v) to 0
for each voxel v
    if(input_texture(v)≠0)compute LIC function output O(v)
```

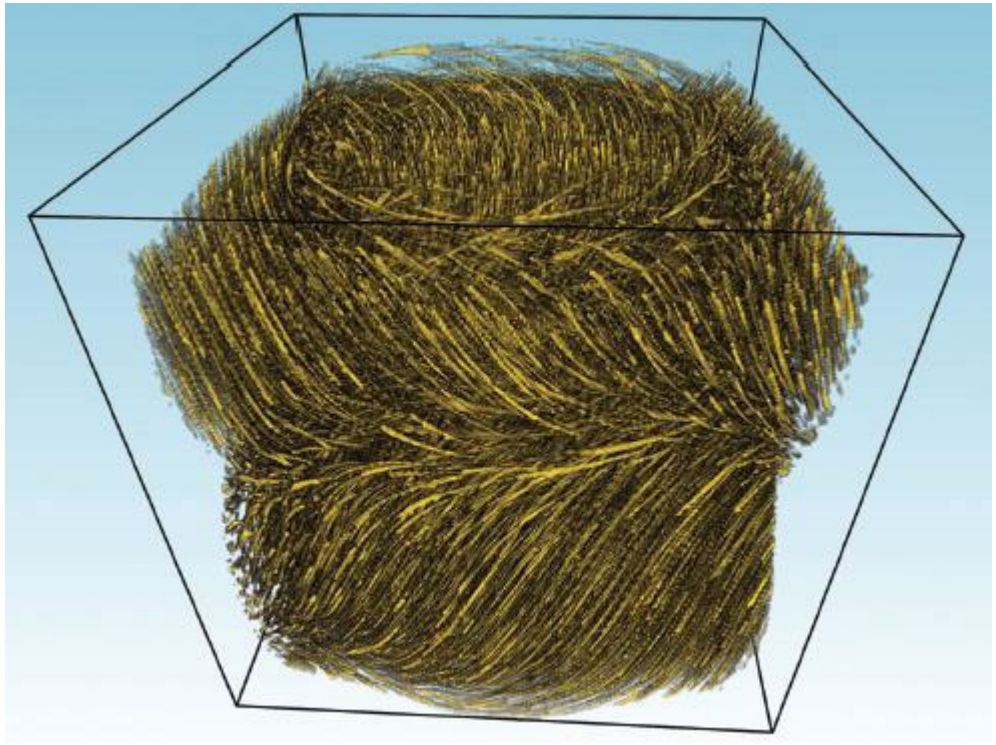


Image Based Flow Visualization

- Image Based Flow Visualization (IBFV) is computationally cheaper than LIC and more general
  - Suitable for dynamic fields such as $O^{\text{vector3}}_{X \times Y \times Z \times T}$
  - Replace expensive per-pixel convolution of LIC by per-pixel advection
  - Repeated advection causes image to fall off-screen
  - Thus, inject time-varying noise image at each frame

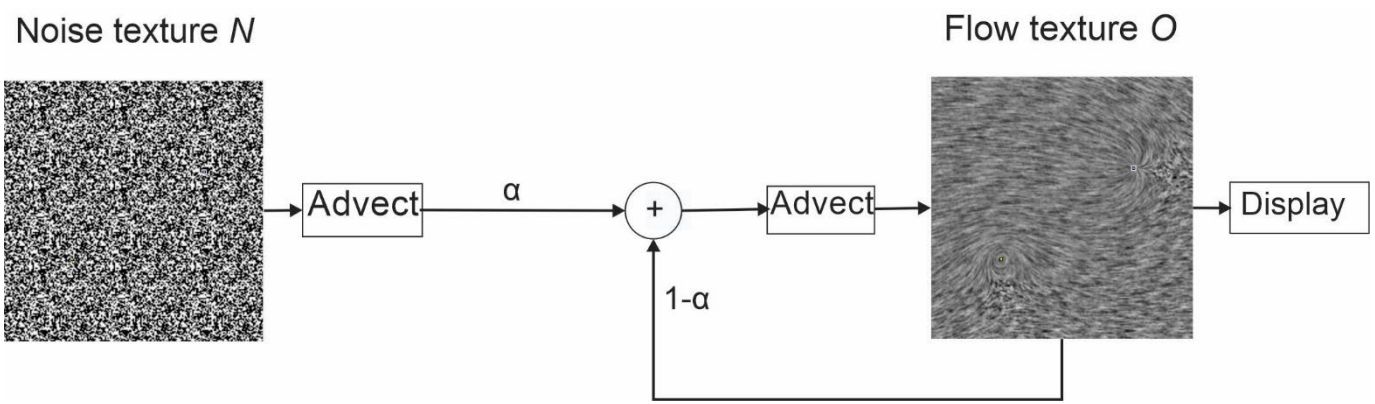


Image Based Flow Visualization (2)

$$O(\mathbf{p}+\vec{D}(\mathbf{p},t)\Delta t, t+\Delta t) = (1-\alpha)O(\mathbf{p},t) + \alpha N(\mathbf{p}+\vec{D}(\mathbf{p},t)\Delta t, t+\Delta t)$$

where

$O(\mathbf{p},t)$: output image value

$N(\mathbf{p},t)$: time-dependent (input) noise image value

$\vec{D}(\mathbf{p},t)$: dynamic vector field value

at pixel **p** and time *t*

$\alpha \in [0,1]$ weighs injection to advection

=1 for pure injection (equals noise image)

=0 for pure advection (with the problem of disappearing texture)

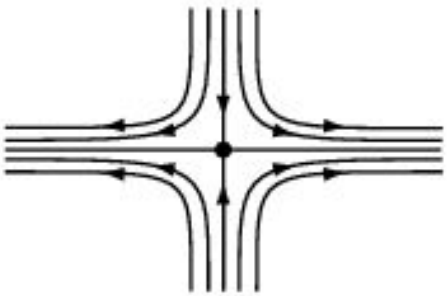
Visualization of Vector Field Topology

- Minimizes the visual clutter from the visualization of vector fields
- Concentrate on important field features (visualize its topology)
- **Critical points**:
  - The points where the field has zero value
  - The most important topological elements
  - Represent singularities of the vector field
  - Classified according to the eigenvalues of the Jacobian matrix of the field evaluated at their position
- For a 2D vector field $\vec{S}(x,y) = \left(S_x(x,y), S_y(x,y)\right)$ the Jacobian matrix is

$$J_2 = \begin{bmatrix} \frac{\partial S_x}{\partial x} & \frac{\partial S_x}{\partial y} \\ \frac{\partial S_y}{\partial x} & \frac{\partial S_y}{\partial y} \end{bmatrix}$$

  - Has 2 eigenvalues that are either both real or both complex

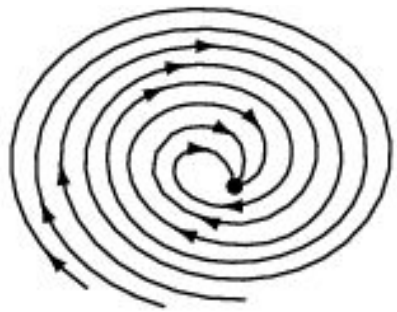
Visualization of Vector Field Topology (2)



Saddle point

$R_1 \cdot R_2 < 0$

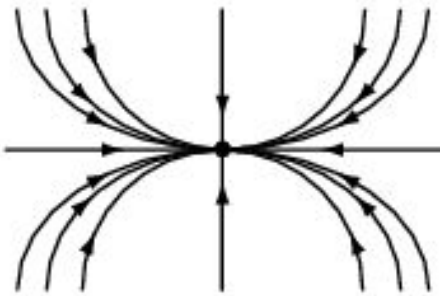
$I_1, I_2 = 0$



Attracting focus

$R_1, R_2 < 0$

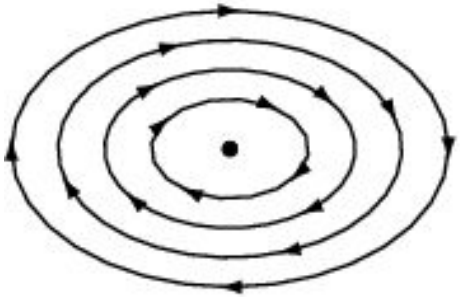
$I_1, I_2 \neq 0$



Attracting node

$R_1, R_2 < 0$

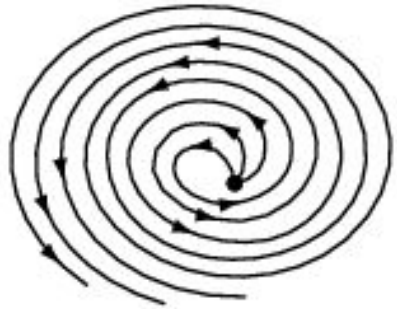
$I_1, I_2 = 0$



Center

$R_1, R_2 = 0$

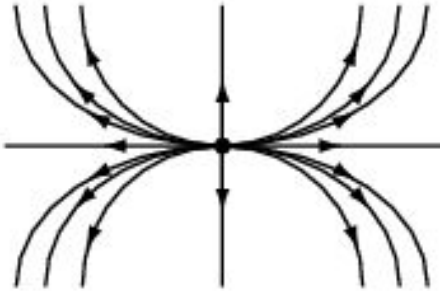
$I_1, I_2 \neq 0$



Repelling focus

$R_1, R_2 > 0$

$I_1, I_2 \neq 0$



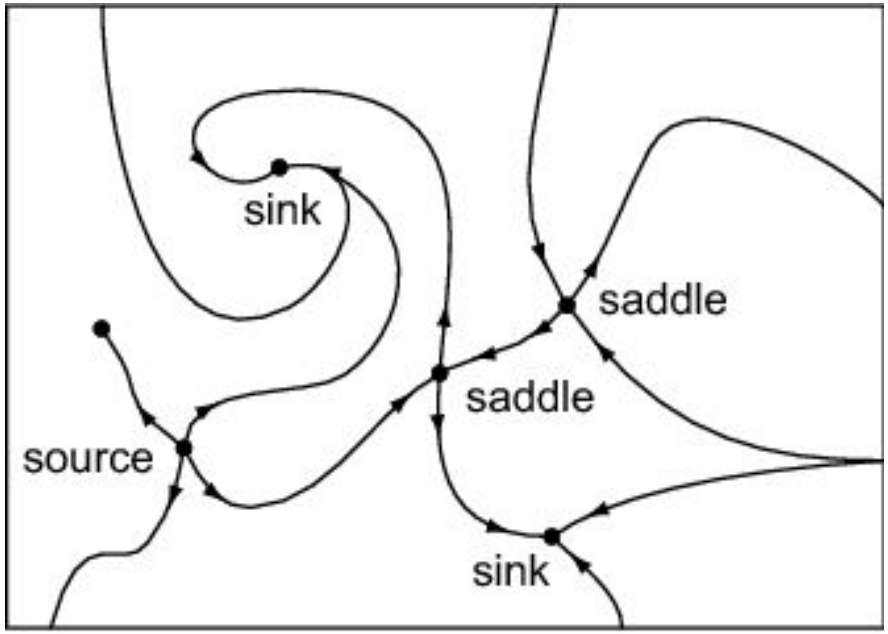
Repelling node

$R_1, R_2 > 0$

$I_1, I_2 = 0$

Visualization of Vector Field Topology (3)

- **Separatrices**
  - Boundaries between field sectors with different type of flow
  - Curves (2D fields) and curves or surfaces (3D fields)
- **Skeleton**
  - A graph of the critical points and the separatrices of a vector field
  - Provides a simplified representation of the field



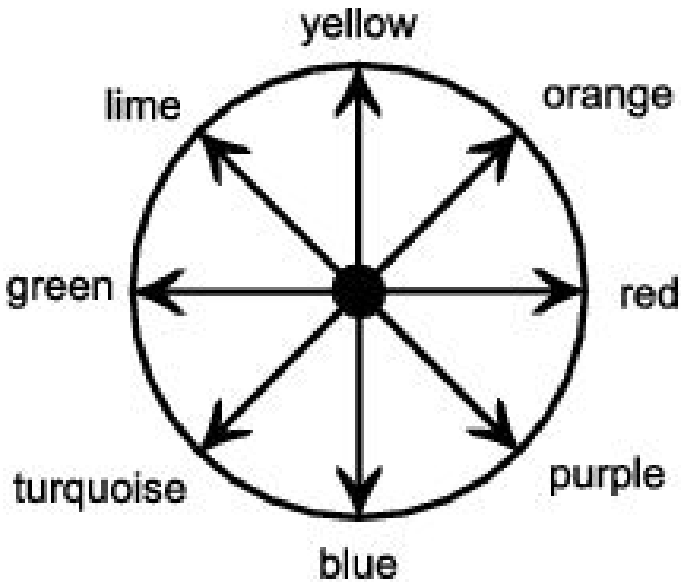
Scalarization

- Simple but lossy
- Discards most components of the vector field creating a scalar field
  - Can be displayed by a simple color-value association
- E.g. pick the magnitude component of the vector field for display and discard all directional information
- Hall's color mapping method
  - Does not discard vector components
  - Maps direction and magnitude to different color characteristics

Scalarization (2)

Hall's method:

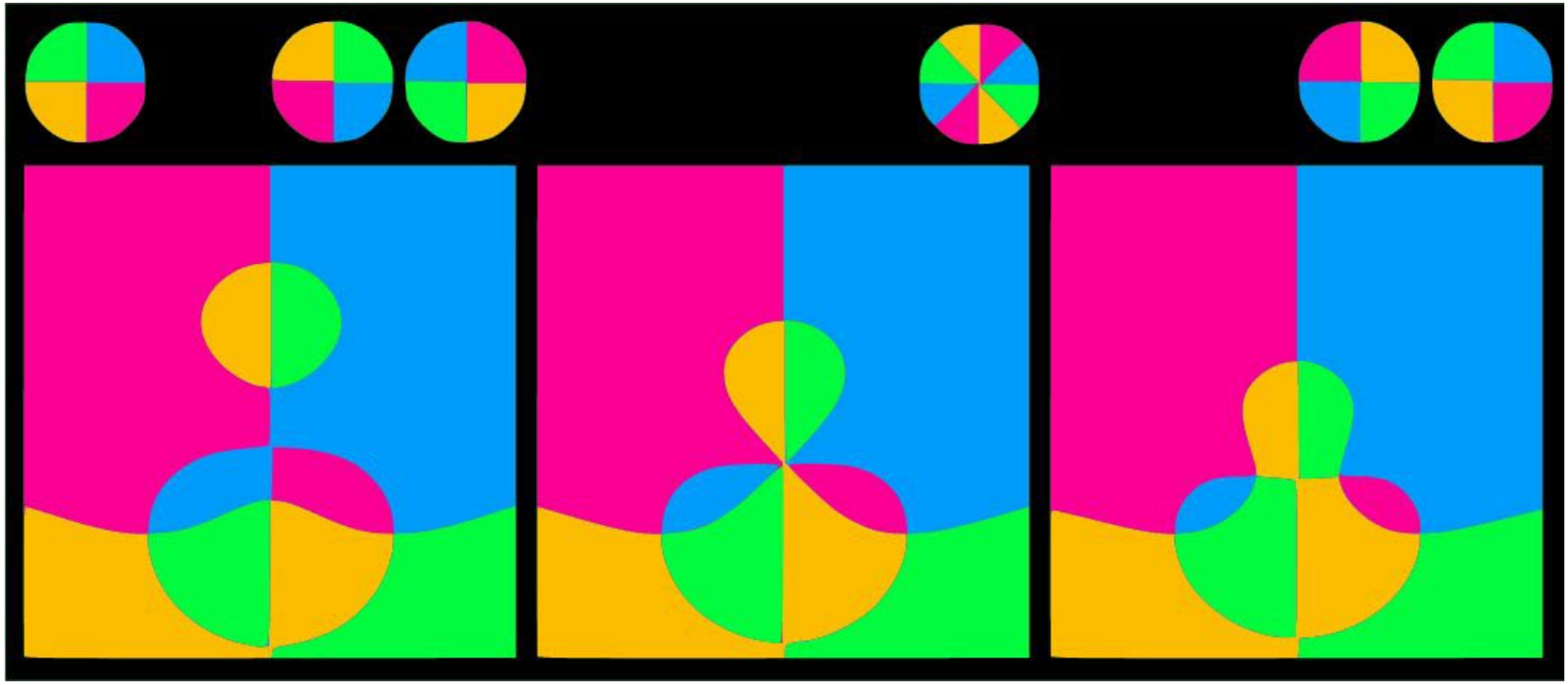
- Assume vectors of $O_{X\times Y}^{\text{vector2}}$ represented in a polar coordinate system as triplets (ρ,θ,φ)
- Map ρ, θ, φ to a spherical color-coordinate system
- θ can represent hue
- φ can represent tone (shade)
- ρ can represent purity



- Thus:
  - pure colors represent vectors of maximum magnitude
  - direction is mapped to hue and tone

Scalarization (3)

- Hard for our brain to associate color with vector characteristics
- Improved if the colors are quantized before display
  - helps us to classify the vectors into a few major direction categories



Vector Field Simplification

- Simplification techniques:
  - Can be used to simplify (reduce) the field before visualizing it
  - Reduction of the number of vectors in a controlled way which aims to preserve important field properties
- The risk of missing important data is reduced
- Generalization of edge-collapse operation:
  - Suitable for vector fields defined over tetrahedral meshes
  - Reduces the tetrahedra that share a collapsed edge to triangles
  - These are deleted
  - Tetrahedra that share a vertex of the collapsed edge must have this vertex updated

Vector Field Simplification (2)

Simplification Algorithm:

- Create a queue of candidate edge-collapses by ordering all edges of the tetrahedral mesh according to an error metric that measures the degradation of the field as a result of each collapse
- Repeat until the simplification target is reached:
  - Remove the edge from the front of the queue
  - Apply the associated edge-collapse
  - Re-compute the error metric for affected queue elements and reorder the queue

Vector Field Visualization Techniques: Overview

	Hedgehogs	Particle advection	Texture based	Vector field topology	Scalarization	Simplification
Full field visualization	x		x		x	
Direction visualization	x	x	x	x	x[2]	x
Magnitude visualization	x	x	x		x[1]	x
No significant information loss	x		x			x
Avoids clutter		x	x	x	x	x

[1] Either vector direction or magnitude is usually visualized with this technique.